

Mini Lesson Sample Script

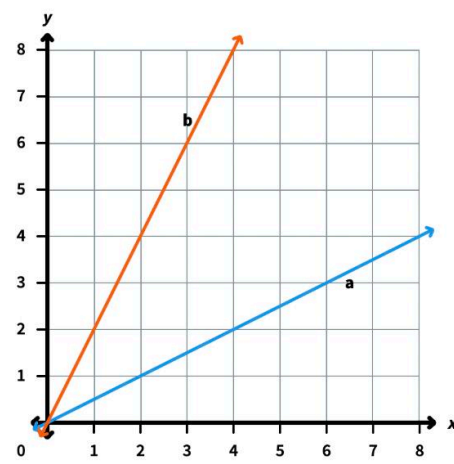
Materials

(S) Activity 1 Worksheet

PROBLEM

- 1 Identify which of two lines on a grid is steeper and connect steepness to the idea of slope as vertical distance divided by horizontal distance.

Two lines are graphed on the grid below. Which line is steeper, line *a* or line *b*?



Answer: Line *b* is steeper.

SAMPLE SCRIPT

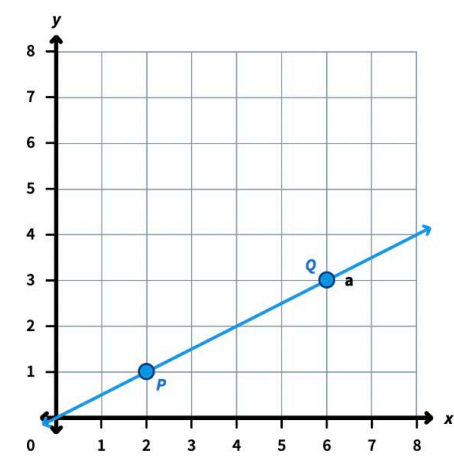
T: Look at the two lines on your worksheet. Imagine each line is a slide at a playground. If you slid down both, which one would you go faster on?

S: Line *b*. It's way more tilted.

T: Right, line *b* is steeper. In math, we have a number that describes how steep a line is. It's called slope. Slope is the vertical distance divided by the horizontal distance between two points on a line. A steeper line has a greater slope. Today we're going to find slopes by picking two points on a line and counting distances on the grid.

- 2 Find the slope of line *a* by choosing two points at grid intersections and calculating the vertical distance divided by the horizontal distance.

Find the slope of line *a*.



Answer: The slope of line *a* is $\frac{2}{4}$, which equals $\frac{1}{2}$.

T: Look at line *a*. I've marked two points, P and Q, where the line crosses grid intersections. Starting at point P, count straight up to the level of point Q. How many units is that?

S: 2 units.

T: That's the vertical distance. Write that down. Now, from there, count straight across to point Q. How many units?

S: 4 units.

T: That's the horizontal distance. So what is the slope? Remember, slope is vertical distance divided by horizontal distance.

S: 2 divided by 4. That's $\frac{2}{4}$.

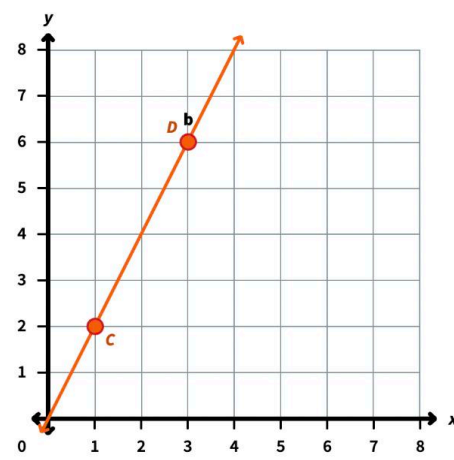
T: Can we simplify that?

S: $\frac{1}{2}$.

T: So the slope of line *a* is $\frac{1}{2}$. That means for every 2 units you go across, the line goes up 1 unit. Write that on your worksheet.

- 3 Determine the slope of line *b* and compare slopes to describe the relative steepness of the two lines.

Find the slope of line *b*. Then explain how the slopes of line *a* and line *b* confirm which line is steeper.



Answer: The slope of line *b* is $\frac{4}{2}$, which equals 2. Line *b* is steeper because its slope of 2 is greater than line *a*'s slope of $\frac{1}{2}$.

T: Now find the slope of line *b* using points C and D. Count the vertical distance and horizontal distance, then calculate the slope. Go ahead.

S: (Work.) The vertical distance from C to D is 4 units. The horizontal distance is 2 units. So the slope is 4 divided by 2, which is 2.

T: Good. So the slope of line *b* is 2. Now, think back to Problem 1. We said line *b* was steeper. Look at the two slopes: line *a* has a slope of $\frac{1}{2}$, and line *b* has a slope of 2. Which slope is greater?

S: 2 is greater than $\frac{1}{2}$.

T: And that matches what we saw — line *b* is steeper. A larger slope means the line goes up more for the same distance across, so it's steeper.